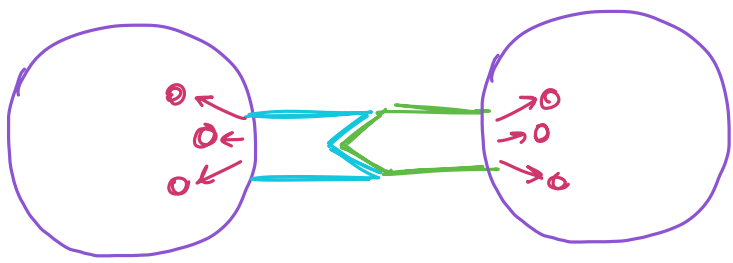


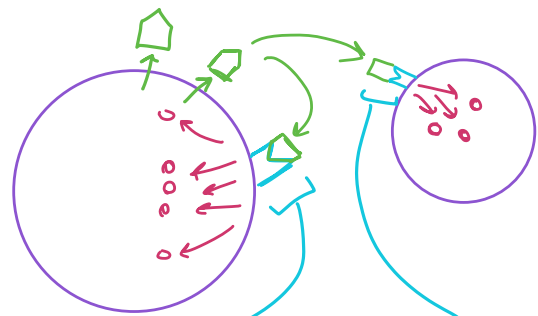
unit 4

- **Signal Reception**—the first stage of cell communication, when the cell receives the signal.
 - ▶ **Ligand**—a signal molecule that produces a response in a cell
 - *** Ligand-binding is REVERSIBLE and NONCOVALENT.
 - ▶ **Target Cell**—cells affected by a particular ligand
 - *** Specificity of the ligand means that only cells with receptors that can bind to the ligand (tertiary structure) will exhibit a response
 - ▶ **Allosteric Regulation**—the protein that the ligand binds to will change shape in another area of the protein as a result of the ligand binding. This will initiate a cellular response.
 - ▶ **Intracellular receptor**—located inside of a cell. Ligands are small or nonpolar and can diffuse through the membrane.
 - **Transcription Factors (Cell Messaging)**—an intracellular receptor can act as a transcription factor to influence gene regulation and expression
 - ▶ **Membrane receptor**—located on the membrane of the cell. Ligands are large or polar and cannot diffuse through the membrane.
 - *** Both ligand-gated ion channels and G-protein linked receptors are examples of membrane receptors.
 - **Ligand-Gated Ion Channel**—channel proteins open and change shape when a ligand binds, which allows certain ions into the cell.
 - **G-Protein Linked Receptor**—a G protein will bind to a membrane receptor once a ligand binds to the receptor and changes its structure. When the G protein binds, a high-energy GTP molecule will bind to the G protein, replacing a low-energy GDP molecule. The G protein will then bind to another membrane protein that initiates a cellular response.
- **Signal Transduction**—the pathway after reception that leads to a cellular response
 - ▶ **Multistep Pathway**—allows for amplification and safer regulation of a signal and response
 - ▶ **Phosphorylation Cascade**—a series of protein kinases phosphorylating other protein kinases, exponentially amplifying the signal, and thus the response.
 - ▶ **Phosphorylation**—attaching a phosphate group to a protein, which typically activates the protein.
 - ▶ **Kinase**—an enzyme that phosphorylates other proteins.
 - ▶ **Dephosphorylation**—removing the phosphate groups from proteins and thus inactivating all previously activated proteins.
 - ▶ **Phosphatase**—a protein that removes phosphate groups from activated proteins to stop a signal
 - ▶ **Secondary Messenger**—small, nonprotein, hydrophilic molecules or ions that move throughout the cell quickly and activate proteins involved in the transduction process
 - ▶ **Cyclic AMP (cAMP)**—one of the most common secondary messengers. cAMP typically activates protein kinase A, which initiates a phosphorylation cascade.

- ▶ **Amplification**—increasing the size and effect of the signal
- ▶ **Signal Specificity**—a receptor protein has a very specific tertiary structure in its active site that requires a very specific ligand with its own specific tertiary structure to bind. Additionally, the proteins that carry out each signal may be different in different cells, meaning that the same ligand can initiate different responses in different cells in the body.
- **Cellular Response**—the final step of cell signaling, the final outcome of a ligand binding. It could be uptake of ions, altering gene expressions, releasing cellular products, increasing reaction rates...
- **Signal Disruption**—either too little of the ligand is produced (Type 1 Diabetes) or the signal receptor is defective and binding either does not occur or does not initiate a response (Type 2 Diabetes).
 - *** Certain poisons also target cell signaling to kill cells.
 - ▶ **Insulin**—a common ligand that initiates uptake of glucose.
 - ▶ **Diabetes**—a condition in which cell signaling is disrupted because of a lack of insulin or a defective receptor.
- **Direct Contact Communication**—cells communicate through direct contact with one another
 - ▶ **Juxtacrine Signaling**—cell surface markers or cell junctions that connect separate cells allows for communication through direct contact.
 - **Plasmodesmata (plant cells)**—gaps that connect plant cells for exchange of signal molecules.
 - **Gap Junctions**—gaps between cells that allow for exchange of signal molecules.
- **Short Distance Communication**—cells are not in direct contact to communicate, but are nearby, using local regulators and signal molecules.
 - ▶ **Autocrine Signaling**—affects the same cells that send the signal. The cell releases ligands that bind to its own receptors.
 - *** Cancer cells can secrete growth signals that bind to their own receptors and cause the cells to divide faster.
 - ▶ **Paracrine Signaling**—chemical signals will spread out and bind to other cells' receptors
 - **Morphogen**—a signal molecule released during embryo development that diffuses and creates a concentration gradient, alerting nearby cells (based on the amount of morphogens they are receiving) how they should develop and begin to differentiate
 - **Quorum Sensing**—bacteria cells secrete signal molecules, and when they receive enough signal molecules from other bacteria, they initiate a response (bioluminescence).
- **Long Distance Communication**—signals released by one cell can travel through the bloodstream to another type of cell
 - ▶ **Endocrine Signaling**—involves secretion of hormones into the blood
 - **Hormone**—a signal molecule that travels through the blood and is secreted by a **gland**
 - **Gland**—organ that secretes particular chemical substances

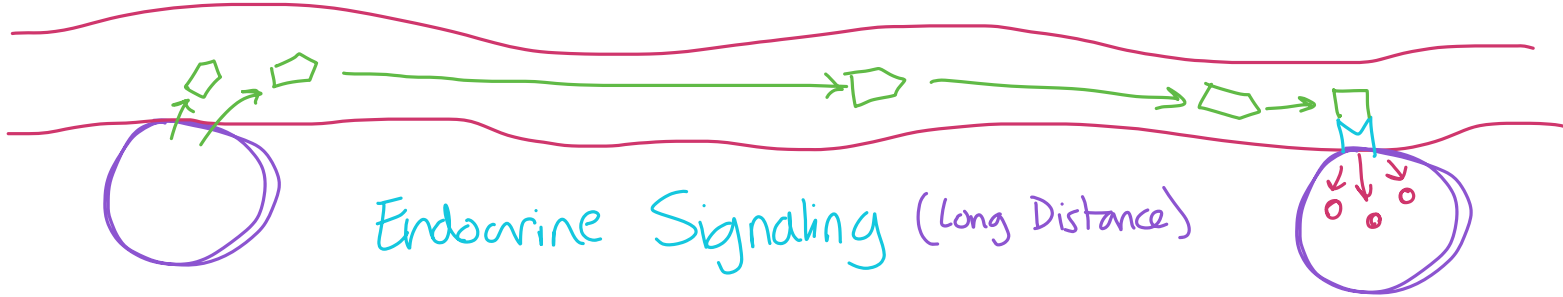


Juxtacrine Signaling
(Direct Contact)



Autocrine Signaling
(Short Distance)

Paracrine Signaling

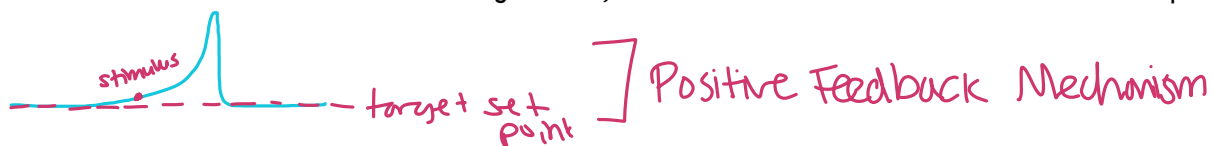


Endocrine Signaling (Long Distance)

- **Homeostasis**—ability of an organism to regulate and retain stable internal conditions despite a changing external environment
- **Negative Feedback Mechanism**—a stimulus initiates a response opposite of what the stimulus was doing to rebalance
- **Target Set Point**—the specific “internal balance” that a feedback mechanism strives to retain



- **Thermoregulation**—an example of a negative feedback mechanism in order to regulate internal temperature
 - **Sweating, Skin Blood Vessel Dilation (Enlargement)**—negative feedback mechanism when temperature is too high
 - **Shivering, Skin Blood Vessel Constriction (Shrink)**—negative feedback mechanism when temperature is too low
- **Stomata**—plant pores in leaves that take in and let out small molecules
 - **Guard Cell Regulation**—an example of negative feedback. When water is scarce, guard cells shrink and close the stomata to ensure no water slips out. When water is available, guard cells expand and open the stomata.
- **Positive Feedback Mechanism**—amplifying the stimulus farther away from the target set point until a favorable outcome is achieved
 - **Ethylene**—a gas that is released by ripening fruit and prompts ripening and in turn, initiates a response from nearby fruit that releases more ethylene, until all the fruit are ripe.



- **Cell Cycle**—the cycling stages of a cell's life, whether it be preparing for division, or dividing, or never dividing again.
 - **Cell Division**—duplication of the cell and its genetic information
 - *** Cell division is important for replacing damaged cells, sexual reproduction, growth, and repair.
 - **Interphase**—cell grows and copies chromosomes to prepare for division (90% of the cell cycle)
 - **G1 (“first gap”) phase**—cell grows in size and doubles its organelles
 - **S (“synthesis”) phase**—cell replicates its DNA and attaches sister chromatids at the centromere
 - **G2 (“second gap”) phase**—cell continues to grow and prepares for division
 - **M (“mitotic”) phase**—cell divides into two identical daughter cells
 - **Mitosis**—genetic information and nucleus divides
 - **Prophase (Prophase+Prometaphase)**—chromatin condenses into chromosomes and centrosomes move to opposite ends of the cell. The nuclear membrane begins to disappear
 - **Centromere**—the point at which two sister chromatids are connected
 - **Kinetochores**—a protein structure at the opposite end of the centromere in each sister chromatid
 - **Sister chromatids**—contain the same genetic information, one chromosome was duplicated during the “S” phase of interphase.
 - **Condensed Chromosome**—a tightly coiled joined group of genetic information
 - **Chromatin**—stringy DNA, uncondensed, and uncoiled
 - **Histones**—proteins around which chromatin is wound
 - **Nucleosome**—a group of chromatin and a histone
 - **Centrosome**—contains two centrioles which create the mitotic spindle
 - **Centriole**—two in each centrosome
 - **Mitotic spindle**—arranges and separates sister chromatids during division
 - **Metaphase**—chromosomes line up along the middle of the cell, attached to the mitotic spindle at kinetochores.
 - **Anaphase**—kinetochores pull each sister chromatid apart and each chromosome sends one chromatid to each end of the cell
 - **Telophase**—nuclear membrane reforms and chromosomes uncondense
 - **Cytokinesis**—cell cytoplasm divides and two identical daughter cells appear
 - **Cleavage Furrow (Animals)**—forms to pinch the cell in two
 - **Cell Plate (Plants)**—forms a piece of the cell wall that will separate the cytoplasm
- **Somatic cell**—diploid body cell
- **Gamete cell**—haploid reproductive cell
- **Diploid**—containing two of each chromosome
 - *** Mitosis results in two diploid somatic cells.
- **Haploid**—containing just one of each chromosome

- **Binary Fission**—in prokaryotes; asexual reproduction; prokaryote's plasma membrane pinches inward and divides the cell after it has replicated its single circular chromosome
- **Cell Cycle Checkpoint**—a point in the cell cycle where go-ahead or stop signals can increase or decrease the speed of the cell cycle.
- **G1 checkpoint**—checks cell growth, available energy and resources, room in the environment for new cells, damage to the DNA, growth factor presence
 - *** The most important in mammalian cells. If this checkpoint is passed, it's likely that the cell will continue to division.
 - **"G0" phase**—the cell will enter a non-dividing stage if it does not pass the G1 checkpoint.
 - **Non-dividing cells**—liver and eye lens cells and other cells never divide or only divide if absolutely necessary (nearby damage) and stay in a constant state of G0
- **G2 checkpoint**—checks that DNA replication happened successfully and that there are ample resources for cell division
 - *** If DNA is damaged, repair will be attempted. If the DNA is beyond repair, the cell will undergo apoptosis.
 - **Apoptosis**—programmed cell death
- **M checkpoint**—ensures the mitotic spindle is correctly assembled and attached to each sister chromatid
- **Cyclin-dependent kinases (CDKs)**—kinases that only initiate a phosphorylation cascade when bound to cyclin, signaling the next stage of the cell cycle. CDKs are always present in a cell.
- **Cyclin**—protein that binds to a cyclin-dependent kinase to activate it. Cyclin numbers fluctuate whereas CDK numbers stay the same.
- **Cyclin-CDK complex**—the only protein complex that can usher in a new period of the cell cycle
- **Stimulation at G1**
 - The presence of growth factors makes G1 cyclin presence increase, which speeds up the cell cycle
- **Inhibition at G1**
 - Cells signal to each other to inhibit CDK-Cyclin complexes when cell density is too high.
- **Stimulation at G2**
 - **Mitosis-Promoting Factor (MPF)**—mitotic cyclins bind to CDKs to bring about mitosis if conditions are favorable
- **Inhibition at G2**
 - **p53**—important protein that inhibits CDK-Cyclin complexes if DNA damage is present and can initiate apoptosis if DNA damage is irreversible.

Proto-oncogenes—code for proteins that promote the cell cycle

Tumor suppressor genes—code for proteins that slow down or inhibit the cell cycle

*** Mutated tumor suppressor genes lead to absence of "stop" proteins, leading to uncontrolled cell division

Oncogenes—a mutated form of a proto-oncogene, which creates proteins that overstimulate the cell cycle

Cancer—overly rapid and harmful cell division

Tumor—a dangerous mass of cancerous cells